

PORT ELIZABETH INTL, South Africa

WMO: 688420

Lat: 33.9864S

Lon: 25.6164E

Elev: 63

StdP: 100.57

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.2	6.8	-0.8	3.5	16.3	1.0	4.1	14.4	15.8	14.7	14.1	14.2	1.9	280	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.6	29.5	18.8	27.6	19.4	26.1	19.7	22.6	25.8	22.0	25.1	21.3	24.3	4.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.8	16.6	24.2	21.0	15.8	23.6	20.2	15.0	22.9	67.2	25.9	64.7	25.3	62.3	24.3	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.5	12.9	11.5	2.0	36.2	1.2	1.8	1.1	37.5	0.3	38.6	-0.4	39.6	-1.3	40.9		
			1.1	24.0	1.4	0.8	0.1	24.5	-0.7	25.0	-1.4	25.4	-2.4	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	21.4	21.7	20.4	18.4	16.6	14.5	14.1	14.6	15.5	17.0	18.4	20.0
	DBStd	3.43	1.89	1.89	2.04	2.59	2.40	2.40	2.45	2.36	2.14	2.31	2.12	1.98
	HDD10.0	1	0	0	0	0	0	0	1	0	0	0	0	0
	HDD18.3	646	2	1	5	30	66	117	134	118	88	53	24	7
	CDD10.0	2811	353	327	323	251	204	136	127	143	166	218	251	311
	CDD18.3	412	96	94	70	31	11	3	3	3	4	13	26	59
	CDH23.3	1558	302	318	207	128	116	64	72	63	28	52	56	151
	CDH26.7	286	24	37	34	45	42	12	15	22	10	18	11	15
Wind	WSAvg	5.3	5.7	5.5	5.3	4.8	4.5	4.5	4.6	5.1	5.6	6.0	6.0	5.7
Precipitation	PrecAvg	599	34	35	51	51	54	57	49	67	50	57	53	39
	PrecMax	1058	120	111	309	147	165	211	215	253	204	213	238	182
	PrecMin	399	5	4	5	3	3	2	1	5	10	10	7	5
	PrecStd	151	27	24	47	35	45	48	45	53	35	38	40	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.0	29.9	30.2	32.1	31.1	28.2	28.8	29.8	27.0	28.2	27.2	28.7
		MCWB	22.1	22.5	21.0	19.4	16.9	14.3	14.8	14.8	16.4	18.4	19.1	21.3
	2%	DB	27.0	27.6	26.9	26.6	26.8	25.1	25.2	24.0	22.2	23.9	24.6	26.0
		MCWB	21.6	21.8	20.7	18.7	16.2	14.1	13.6	14.1	16.4	17.9	19.1	20.5
	5%	DB	26.0	26.2	25.2	23.9	22.9	22.0	21.8	20.5	20.8	22.0	23.2	24.9
		MCWB	20.9	21.0	20.0	18.6	15.8	13.4	13.0	14.5	16.1	17.4	18.2	19.9
	10%	DB	25.0	25.1	24.1	22.2	20.9	19.7	19.1	19.0	19.6	20.9	22.1	23.9
		MCWB	20.3	20.5	19.4	17.7	15.9	13.6	13.3	14.5	15.4	16.7	17.7	19.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.6	23.7	22.6	21.2	19.4	17.3	16.7	17.8	18.6	20.0	21.1	22.7
		MCDB	26.9	27.4	26.6	26.9	23.2	20.8	21.2	21.7	22.1	24.9	24.6	25.5
	2%	WB	22.5	22.7	21.6	20.1	18.0	16.0	15.7	16.7	17.7	18.8	20.1	21.6
		MCDB	25.4	25.9	25.1	24.1	22.0	19.9	19.8	19.5	20.7	22.2	23.0	24.5
	5%	WB	21.8	22.1	20.8	19.4	17.4	15.2	15.1	15.9	16.9	18.1	19.3	20.7
		MCDB	24.7	25.0	24.0	22.7	21.0	19.2	18.8	18.8	19.6	21.0	22.1	23.7
	10%	WB	21.1	21.4	20.2	18.6	16.6	14.5	14.3	15.2	16.1	17.4	18.5	20.0
		MCDB	24.0	24.1	23.1	21.5	20.0	18.4	17.9	18.2	18.7	20.0	21.2	22.7
Mean Daily Temperature Range	5% DB	MDBR	7.4	7.6	7.8	8.8	10.1	11.4	11.2	10.0	8.7	7.9	7.7	7.6
		MCDBR	8.4	8.9	9.8	11.9	15.0	16.8	17.4	14.2	11.2	9.8	9.2	8.9
		MCWBR	4.2	4.4	4.7	5.5	7.2	8.1	8.2	7.4	6.3	5.3	4.9	4.5
	5% WB	MCDBR	7.4	8.1	8.5	10.0	11.1	12.2	12.2	10.8	9.6	8.9	8.3	8.1
		MCWBR	4.0	4.2	4.4	5.2	6.4	7.4	7.2	6.5	6.1	5.1	4.8	4.4
Clear-Sky Solar Irradiance	taub		0.389	0.392	0.377	0.353	0.339	0.327	0.329	0.367	0.385	0.399	0.380	0.384
	taud		2.470	2.471	2.496	2.536	2.548	2.555	2.547	2.431	2.373	2.368	2.457	2.472
	Ebn at Noon		949	924	902	868	826	811	828	837	877	907	951	957
	Edh at Noon		117	113	103	88	77	72	76	96	114	124	118	118
All-Sky Solar Radiation	RadAvg		6.95	6.22	5.20	3.92	2.98	2.61	2.79	3.59	4.76	5.73	6.62	7.19
	RadStd		0.36	0.29	0.34	0.20	0.14	0.15	0.18	0.22	0.28	0.42	0.45	0.45

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.38	+0.69	+0.84	N/A	N/A	N/A	N/A	-85	+142	+45	
(47)	Regional (5 neighbors)	+0.38	+0.69	+0.84	N/A	N/A	N/A	N/A	-85	+142	+45	

Nomenclature: See separate page